

Project Report

University Parking Network

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1. Introduction

Parking congestion in the university is a growing issue, leading to harassment among students, staff, and visitors. Inefficient parking systems can result in wasted time and increased vehicle emissions. To address this challenge, an intelligent parking management system is proposed. This system leverages advanced network design and simulation tools to provide an organized and efficient solution for managing parking spaces. By integrating modern networking techniques, the system aims to enhance accessibility, streamline traffic flow, and optimize space utilization.

2. Objective

The objective of our project is to design and implement an intelligent parking management system for our university. The goals include:

- Minimizing parking congestion and improving traffic flow within the campus.
- Enhancing the user experience by providing a structured and efficient parking solution.
- Implementing a scalable network infrastructure to support real-time monitoring and control of parking spaces.
- Ensuring reliable communication and integration of devices such as laptops, printers, and servers to facilitate system operations.

3. Methodology

We used Cisco Packet Tracer, is a tool built by Cisco. This tool provides a network simulation to practice simple and complex networks. **Workspace:**

1. Logical

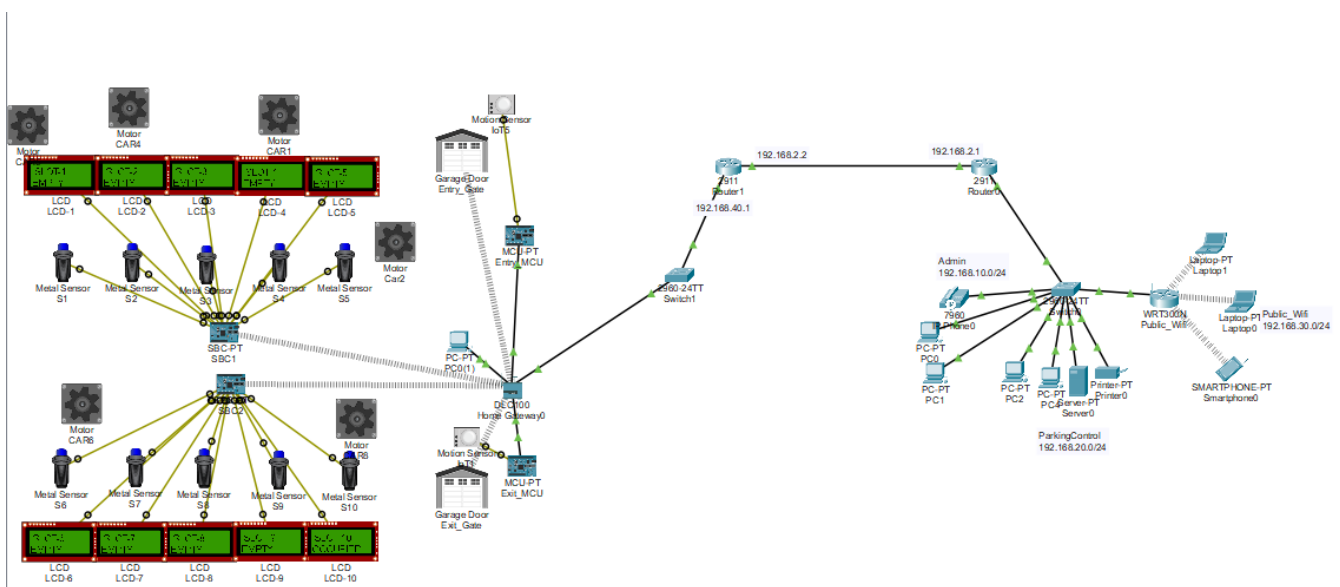
The logical workspace shows the logical network topology of the network the user has built. It represents the placing, connecting and clustering virtual network devices.

2. Physical

The physical workspace shows the graphical physical dimension of the logical network. It depicts the scale and placement in how network devices such as routers, switches and hosts would look in a real environment. It also provides geographical representation of networks, including multiple buildings, cities and wiring closets.

Network Design and Simulation:

- The project will use **Cisco Packet Tracer** to design and simulate the parking management system.
- The network will include interconnected routers, switches, and end devices such as PCs, laptops, printers, and IP phones.
- **Infrastructure Setup:**
 - Multiple subnets will be created for different network sections to ensure efficient traffic management and device communication.
 - VLANs will be implemented to segment traffic for different purposes, such as Parking Control, Public WIFI, and administration.
- **Server Configuration:**
 - A central server will be set up to provide essential services, including **DHCP**, **Email**, and **HTTP**.
- **Device and Communication Configuration:**
 - Routers and switches will be configured to ensure reliable communication across the network.
- **Simulation and Testing:**
 - The system will be tested using simulated scenarios to validate its ability to reduce congestion, optimize parking, and ensure seamless communication between devices.

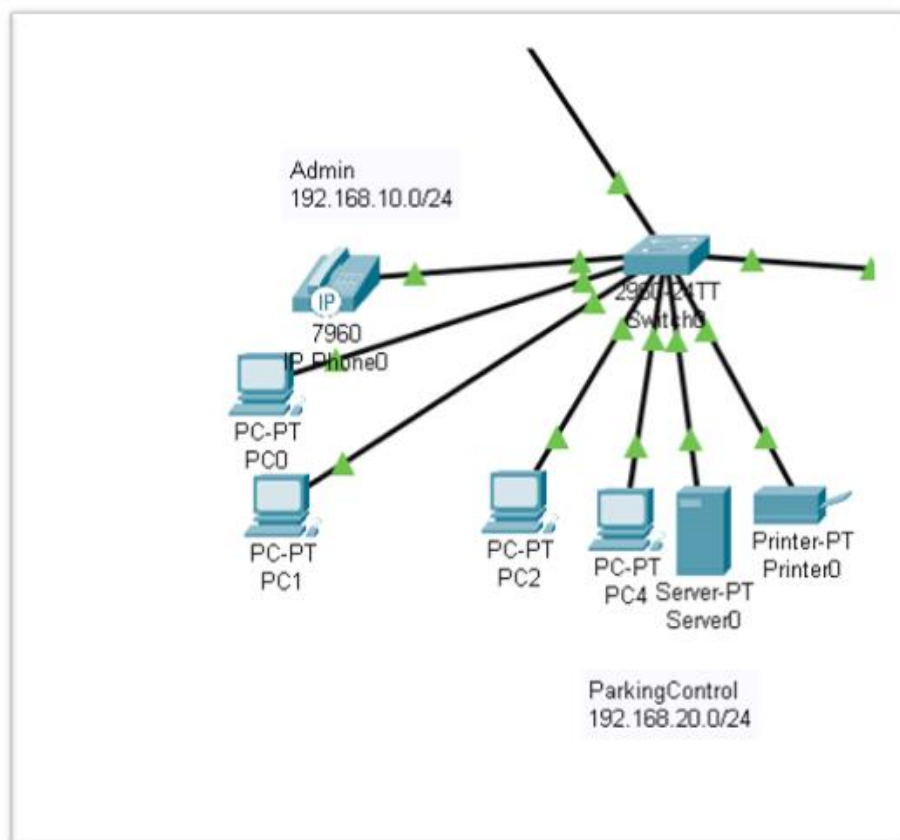


Connection Setup:

- **Connection Setup:**

Device	Function
Router	Connect LANs and facilitate communication between different networks
PC	Used as workstations for each department to do a specific operation
Switch	Provide connectivity between devices in the LAN
Wireless Router	Wireless connectivity between devices in the LAN
Server	Host services for ex: DHCP, Mail, HTTP servers
IP Phone	Voice communication between individuals

- **Implementation and Devices Configuration:**



Admin Part:

Admin and Parking Control sections are organized to manage administrative tasks and parking control efficiently.

VLAN Admin

IP Address: 192.168.10.0/24 Gateway: 192.168.10.1

Subnet Mask: 255.255.255.0

Devices	Gateway	IP
PC0	192.168.10.1	192.168.10.2/24
PC1	192.168.10.1	192.168.10.3/24
Phone0	192.168.10.1	-

- (PC0) and an IP Phone (7960) are connected to the switch.
- The IP Phone is likely used for communication between admin staff.

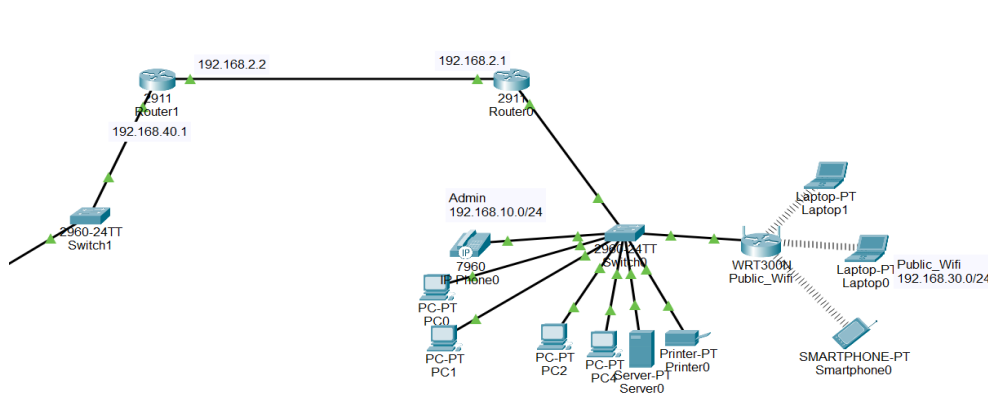
VLAN Parking Control

IP Address: 192.168.20.0/24 Gateway: 192.168.20.1

Subnet Mask: 255.255.255.0

- Ensures real-time monitoring and control of the parking system.
- Provides a centralized platform for managing parking spots, sensors, and possibly billing or access logs.
- Supports efficient communication between devices for parking automation and problem-solving.

Devices	MAC	IP	Gateway
PC2	0090.0CE9.EBD4	192.168.20.3/24	192.168.20.1
PC4	0002.168B.E828	192.168.20.6/24	192.168.20.1
Server	0001.42CE.B721	192.168.20.2/24	192.168.20.1
Printer	0009.7CBC.E721	192.168.20.7/24	192.168.20.1



Second part:

This network system connects multiple networks (VLANs) and allows smooth communication. Switches help connect local devices within the network and link to routers for additional communication.

Public Wi-Fi

IP Address: 192.168.30.0/24

Gateway: 192.168.10.1

Subnet Mask: 255.255.255.0

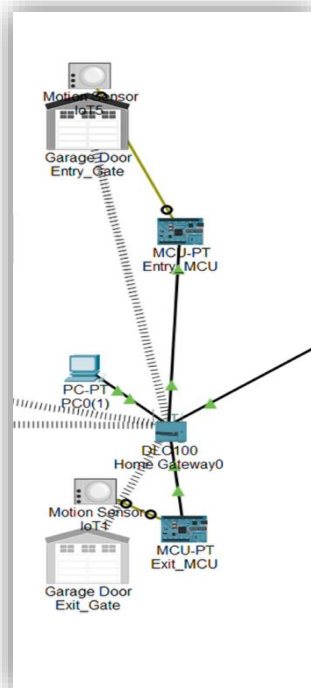
Devices	MAC	IP
Laptop0	0005.5E45.3573	192.168.5.102/24
Laptop1	0090.2BB2.91BC	192.168.5.103/24
Smartphone0	0090.2BB2.91BC	192.168.5.100/24

- Laptop 0, Laptop 1, and Smartphone 0 are connected to Public Wi-Fi
- Public Wi-Fi provides wireless access, allowing visitors to connect to the networks or the internet safely using a username and password.

• Routers

Devices	MAC	IP	Gateway	Subnet Mask
Router0	00D0.9799.6E01	192.168.2.1/24	192.168.2.1	255.255.255.0
Router1	00D0.D3B8.8E01	192.168.2.2/24	192.168.2.2	255.255.255.0

Routers are used to route traffic between the internal networks (VLANs), Public Wi-Fi, and the internet.



Third Part:

This network is designed to automate the monitoring and management of parking slots. It comprises several interconnected components, including metal sensors, LCD displays, motors, and a central SBC (Single Board Computer) that manages communication and operations. The network enables real-time updates of slot statuses and automates processes like barrier control.

Public Wi-Fi:

Public Wi-Fi provides wireless access, allowing visitors to connect to the networks or the internet safely using a username and password.

Network	IP Address	Default Gateway	Subnet Mask
Public Wi-Fi	192.168.30.0/24	192.168.10.1	255.255.255.0

Devices connected to Public Wi-Fi:

Device	MAC Address	IP Address
Laptop0	0005.5E45.3573	192.168.5.102/24
Laptop1	0090.2BB2.91BC	192.168.5.103/24
Smartphone0	0090.2BB2.91BC	192.168.5.100/24

Routers

Routers are used to route traffic between the internal networks (VLANs), Public WiFi, and the internet.

Device	MAC Address	IP Address	Gateway	Subnet Mask
Router0	00D0.9799.6E01	192.168.2.1/24	192.168.2.1	255.255.255.0
Router1	00D0.D3B8.8E01	192.168.2.2/24	192.168.2.2	255.255.255.0

Network VLANs

The network is divided into multiple VLANs for efficient segmentation and communication.

VLAN Admin

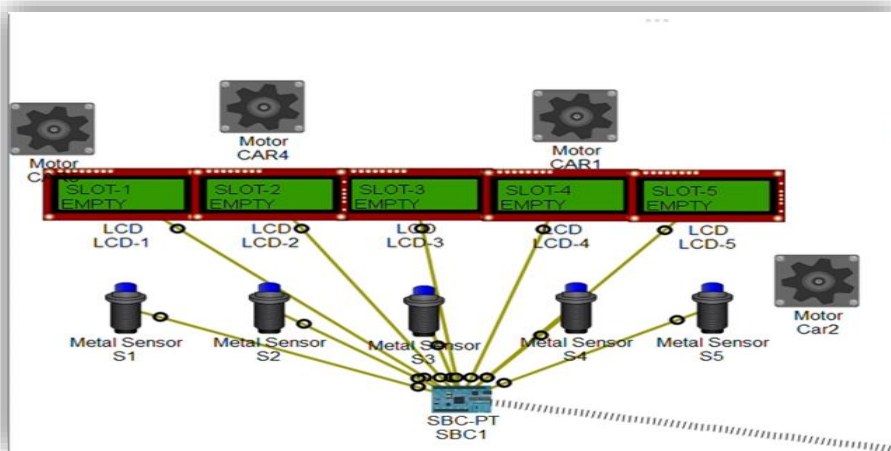
IP: 192.168.10.0/24 | Default Gateway: 192.168.10.1 | Subnet Mask: 255.255.255.0
 Purpose: Enables administrative staff to manage network devices and facilitate office operations.

VLAN Parking Control

IP: 192.168.20.0/24 | Default Gateway: 192.168.20.1 | Subnet Mask: 255.255.255.0
 Purpose: Ensures real-time monitoring and control of the parking system, including managing sensors and logs.

Devices connected to VLAN Parking Control:

Device	MAC Address	IP Address	Purpose
PC2	0090.0CE9.EBD4	192.168.20.3/24	Monitors parking system activities.
PC4	0002.168B.E828	192.168.20.6/24	Facilitates additional monitoring tasks.
Server	0001.42CE.B721	192.168.20.2/24	Centralized data management and log storage.
Printer	0009.7CBC.E721	192.168.20.7/24	Prints parking logs and other administrative documents.



Forth Part:

The Parking Slot Monitoring Network is designed to automate the management and monitoring of parking spaces. It utilizes a centralized SBC (Single Board Computer) to process data from metal sensors and display the status on LCD screens. Motors are used to control physical barriers or other automated systems in the parking area.

SBC Description:

The SBC (single-board computer) is the central hub for managing sensors and LCDs. It communicates with connected devices to monitor parking slots and display their statuses.

Details of the SBC:

Device	IP Address	Wireless Signal Strength	MAC Address
SBC1	169.254.35.189/16	80%	0010.1134.E4D8

Metal Sensors

The metal sensors detect the presence of vehicles in parking slots. Each sensor corresponds to a specific slot and relays the data to the SBC.

Details of the Metal Sensors:

Sensor	Physical Location
S1	Intercity > Home City > Corporate Office > S1
S2	Intercity > Home City > Corporate Office > S2
S3	Intercity > Home City > Corporate Office > S3
S4	Intercity > Home City > Corporate Office > S4

S5	Intercity > Home City > Corporate Office > S5
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LCD Displays:

The LCDs show the status of each parking slot (e.g., empty or occupied) based on data received from the SBC.

Details of the LCD Displays:

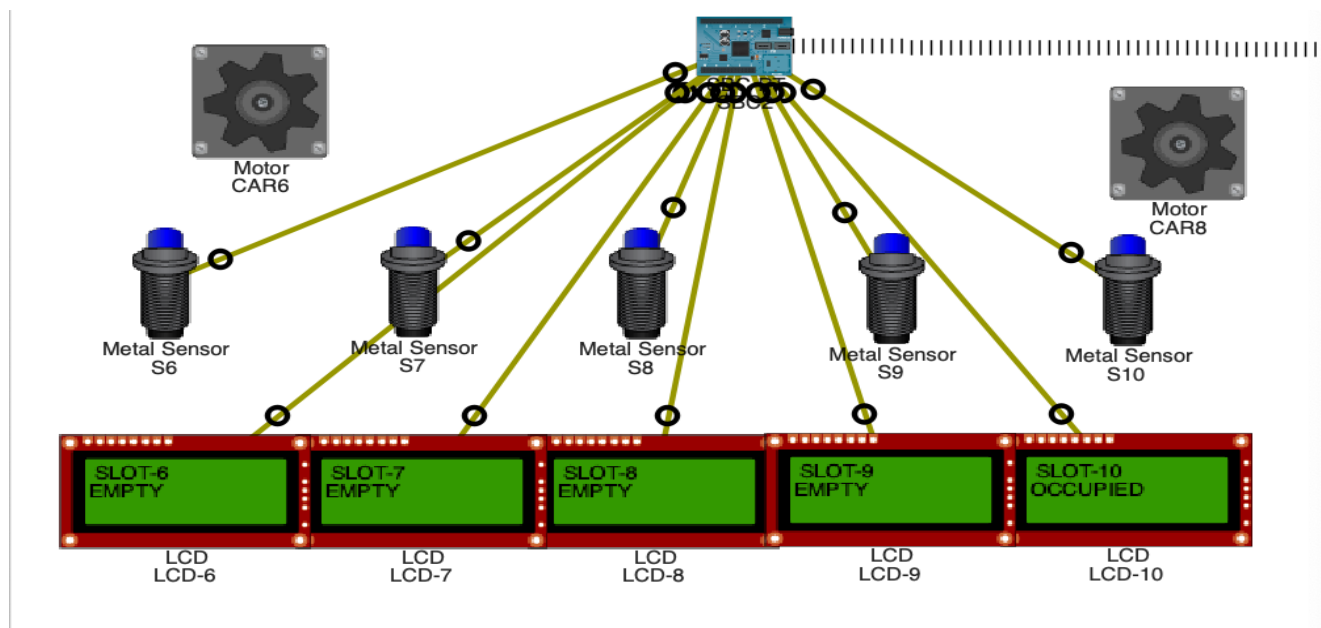
LCD	Physical Location
LCD-1	Intercity > Home City > Corporate Office > LCD-1
LCD-2	Intercity > Home City > Corporate Office > LCD-2
LCD-3	Intercity > Home City > Corporate Office > LCD-3
LCD-4	Intercity > Home City > Corporate Office > LCD-4
LCD-5	Intercity > Home City > Corporate Office > LCD-5

Motors:

The motors are used to automate parking processes, such as opening or closing barriers. Each motor is associated with a specific parking slot.

Details of the Motors:

Motor	Physical Location
CAR1	Intercity > Home City > Corporate Office > CAR1
CAR2	Intercity > Home City > Corporate Office > CAR2
CAR4	Intercity > Home City > Corporate Office > CAR4



Last part:

The Parking Slot Monitoring Network automates the management of parking spaces by integrating sensors, LCD displays, and motors, all centrally managed by an Arduino board. This system ensures real-time monitoring and status updates of parking slots.

Device: SBC2:

SBC2 (Single Board Computer 2) is a key device in the system, acting as a controller for managing parking-related data. It connects wirelessly with sensors, LCDs, and possibly motors to process and distribute data efficiently.

Device Details:

Device Model: SBC-PT (Packet Tracer Single Board Computer model).

Wireless Interfaces:

Wireless3: Active and functioning ("Up").

Bluetooth: Active and functioning ("Up").

Role of SBC2 in the System:

Central Controller:

SBC2 is responsible for processing data from metal sensors (e.g., S6-S10) and coordinating with LCDs (e.g., LCD-6 to LCD-10) to display the real-time status of parking slots.

Wireless Communication:

The SBC uses wireless interfaces to receive input from sensors and send output to other devices. Bluetooth might be used for specific short-range communication, such as connecting with motors or nearby devices.

Networking:

While SBC2 does not currently have a configured gateway or DNS server, it operates on a local network with IPv4 addressing (169.254.x.x/16 range). This range is commonly used for devices on the same LAN when no DHCP server is present.

Integration with Parking System:

Input: Collects data from sensors detecting vehicle presence.

Output: Updates parking slot statuses on LCDs and potentially signals motors to open/close barriers.

Metal Sensors:

The metal sensors are responsible for detecting the presence of vehicles in parking slots. Each sensor is physically wired to the Arduino (or SBC) and corresponds to a specific parking slot.

Details of the Metal Sensors:

Sensor	Parking Slot	Physical Location
S6	Slot 6	Intercity > Home City > Corporate Office > S6
S7	Slot 7	Intercity > Home City > Corporate Office > S7
S8	Slot 8	Intercity > Home City > Corporate Office > S8
S9	Slot 9	Intercity > Home City > Corporate Office > S9
S10	Slot 10	Intercity > Home City > Corporate Office > S10

LCD Displays:

The LCD screens display the real-time status of parking slots (e.g., **EMPTY** or **OCCUPIED**) based on input from the Arduino/SBC. Each display is assigned to a specific parking slot.

Details of the LCD Displays:

LCD	Parking Slot	Physical Location
LCD-6	Slot 6	Intercity > Home City > Corporate Office > LCD-6
LCD-7	Slot 7	Intercity > Home City > Corporate Office > LCD-7
LCD-8	Slot 8	Intercity > Home City > Corporate Office > LCD-8

LCD	Parking Slot	Physical Location
LCD-9	Slot 9	Intercity > Home City > Corporate Office > LCD-9
LCD-10	Slot 10	Intercity > Home City > Corporate Office > LCD-10

Motors:

The motors automate physical actions in the parking system, such as opening or closing barriers or moving vehicles to specific slots. Each motor corresponds to a specific parking slot.

Details of the Motors:

Motor	Parking Slot	Physical Location
CAR6	Slot 6	Intercity > Home City > Corporate Office > CAR6
CAR8	Slot 8	Intercity > Home City > Corporate Office > CAR8

NOTE:

"The metal sensors, LCDs, and motors in the system are directly connected to the Arduino/SBC via physical wiring. As such, they do not have IP or MAC addresses, but their functionality and physical locations are detailed in the tables above."